ALGEBRA 1 · QUADRATIC FUNCTIONS AND EQUATIONS

# Parabola Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Parabola Explorer** on Mr. K's Classroom: Algebra 1 → Quadratic Functions and Equations → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

## 1 Predict before you touch the tool

1. What do you think a negative value of a does to $y = ax^2$?

   **It flips the parabola so it opens down.**

2. Compared with $y = x^2$, which way does $y = (x - 3)^2$ move?

   **Right 3 units.**

## 2 Explore use the tool and record what you see

**Vertex form $y = a(x - h)^2 + k$**

| Equation | Vertex | Opens | Compared with $y = x^2$ |
|---|---|---|---|
| $y = x^2 + 4$ | **(0, 4)** | **up** | **up 4** |
| $y = (x - 3)^2$ | **(3, 0)** | **up** | **right 3** |
| $y = (x + 2)^2 - 5$ | **(–2, –5)** | **up** | **left 2, down 5** |
| $y = -x^2$ | **(0, 0)** | **down** | **reflected** |
| $y = 3x^2$ | **(0, 0)** | **up** | **narrower** |
| $y = \frac{1}{2}(x - 1)^2 + 2$ | **(1, 2)** | **up** | **wider, right 1, up 2** |

**Standard form $y = ax^2 + bx + c$**

Use the Standard form tab.

| Equation | Discriminant $b^2 - 4ac$ | Number of x-intercepts |
|---|---|---|
| $y = x^2 - 2x - 3$ | **16** | **2** |
| $y = x^2 - 4x + 4$ | **0** | **1** |
| $y = x^2 + 2x + 5$ | **–16** | **0** |
| $y = -x^2 + 6x - 5$ | **16** | **2** |

## 3 Write the rule fill in each blank from what you observed

1. k moves the parabola **up** or **down**.
2. h moves the parabola left or right, in the **opposite** direction of its sign in $(x - h)$.
3. If $a < 0$ the parabola opens **down**. If $|a| > 1$ it is **narrower** than $y = x^2$.
4. The vertex of $y = a(x - h)^2 + k$ is **(h, k)**.

5. If $b^2 - 4ac > 0$ there are **2** x-intercepts; if it equals 0 there is **1**; if it is negative there are **0**.

## 4 Check no tool: show your work

1. Give the vertex of $y = -2(x + 4)^2 + 1$. Which way does it open?

   **(−4, 1); down**

2. Write $y = x^2$ shifted left 3 and down 6.

   **$y = (x + 3)^2 - 6$**

3. How many x-intercepts does $y = 2x^2 + 3x + 5$ have?

   **None: $9 - 40 = -31$**

4. Find the axis of symmetry of $y = x^2 - 8x + 3$.

   **$x = 4$**

## 5 Reflect

Why does $(x - 3)^2$ move the graph right, not left?

**The vertex is where $x - 3 = 0$, which is $x = 3$, to the right of 0.**